Version Control

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What is version control?

Version control is a system that records and controls the changes made in a file or project in a way that the user can access its older versions and also see when each change was made. It is an important system in software programming as changes are constantly made in the source code by a programmer.

Why is version control important?

A software project has many files in which many changes are made over time by different people working on the project.

It helps track changes because we can find out:

- What was changed?
- When was it changed?
- Who made the change?
- Why was it changed?

It allows to access older versions of the project:

Suppose a recent change in programme shows an error or a part of the programme was accidently deleted, and we need to restore the previous version that was working properly. This can be done using version control.

It allows multiple coders to collaborate:

Developers can work on the same project at the same time. Version control help them combine their work and manage changes made by different people.

It makes debugging easier:

A coder can check the history of the code to identify when a bug was introduced and can fix the programme efficiently.

It allows developers to experiment:

New features can be tried on the code without the fear of losing the original code.

State and explain different types of version control?

There are three types of version control, local version control, centralized version control and distributed version control.

- **Local Version Control System:**

Here all the versions of the project are kept on one computer. It maintains all the versions of the project instead of relying on manually named copies. It does not require network and is easier for individual use. But since it can be accessed on only one computer, it is not ideal for a tam project and damage to the computer can also lose the repository.

- **Centralized Version Control System:**

Here a central server stores the main repository and developers connect to this server to access the project. Since everyone can work on the common project, it is ideal for team projects. It also allows admins to control access to the project making it easy to manage centrally. Since it relies on one server, a server crash can affect the project and some operations might even be unavailable. Giving access every time for many operations can slow down and restrict a programmers work and cause delays.

- **Distributed Version Control System:**

Here all the programmers have the repository along with its history. They can make changes to the repository with connecting to the central server. These changes can later be combined with other repositories. This provides flexibility to distributed teams.

Mention the important git commands and explain them

git init – Initialize a repository.

git clone – Copy a remote repository.

git add – Stage changes.

git commit – Save changes in the repository.

git status – Check the status of files.

git log – View commit history.

git branch – Create or view branches.

git merge – Combine branches.

git pull – Get and integrate remote changes.

git push – Upload local changes to a remote repository.